

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and IBM Bob”

Project Tile – [Research Agent]

Domain of Project – [Education]

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Recommended font style and size:

Font Style: Arial

Font Size (Heading): 28

Font Size (Content): 20

Problem Statement -

Problem Statement : Research Agent

The Challenge

Research is often time-consuming and difficult because individuals struggle with **finding reliable information, understanding large amounts of data, reviewing research papers, and identifying relevant insights**. Information is scattered across different websites, journals, and documents, making it difficult to verify sources and stay updated. Without **personalized, real-time research assistance**, users may spend hours searching, comparing, and summarizing information manually.

The Objective

Build an **AI-powered Research Agent** that delivers **personalized, reliable, and efficient research assistance** for students, researchers, professionals, and organizations. The solution should:

- **Intelligent Research Assistance** – Understand research queries and identify relevant information from reliable sources.
- **Research Data RAG Retrieval** – Fetch, retrieve, and summarize information from research papers, journals, articles, and trusted databases.
- **Paper Analysis & Summarization** – Analyze research papers and provide concise summaries, key findings, methodologies, and conclusions.
- **Source Verification** – Identify credible sources, compare information, and highlight potentially unreliable or conflicting information.
- **Multi-Agent System** –
 - **Research Knowledge Agent** – Finds and retrieves relevant research information.
 - **Paper Analysis Agent** – Reads and analyzes research papers and documents.
 - **Fact-Checking Agent** – Verifies claims and cross-checks information from multiple sources.
 - **Research Recommendation Agent** – Suggests relevant papers, topics, and research directions.
- **Research Dashboard** – Display collected sources, key findings, research summaries, citations, and progress in an organized manner.

Proposed Solution -

Proposed Solution: AI-Powered Research Agent

The proposed solution is a **multi-agent AI-powered Research Agent** built using **IBM Bob, IBM watsonx.ai, and Granite Models** to provide personalized, reliable, and efficient research assistance. The system addresses challenges in information retrieval, research paper analysis, source verification, and knowledge discovery through an intelligent, modular architecture.

At its core, the solution uses a **Retrieval-Augmented Generation (RAG)** pipeline to fetch accurate and relevant information from trusted sources such as research papers, academic journals, publications, and reliable databases. The retrieved data is embedded and stored in a vector database, enabling the system to efficiently search, retrieve, and summarize relevant research information using Granite models.

The system consists of four key agents:

Research Knowledge Agent retrieves and summarizes relevant information from research papers, journals, and trusted sources using RAG.

Paper Analysis Agent analyzes research papers and documents to extract key findings, methodology, results, and conclusions.

Fact-Checking Agent verifies claims by cross-checking information across multiple reliable sources and identifies conflicting or unsupported information.

Research Recommendation Agent suggests relevant research papers, topics, keywords, and potential research directions based on the user's query and interests.

The solution includes a **user-friendly dashboard** (built with Streamlit or React) that displays research results, paper summaries, key findings, sources, citations, and research progress in an organized manner.

IBM Bob is used to **orchestrate agent workflows**, while watsonx.ai ensures scalable model deployment, embeddings, and AI governance. **Granite models** power reasoning, summarization, information extraction, and personalized research assistance.

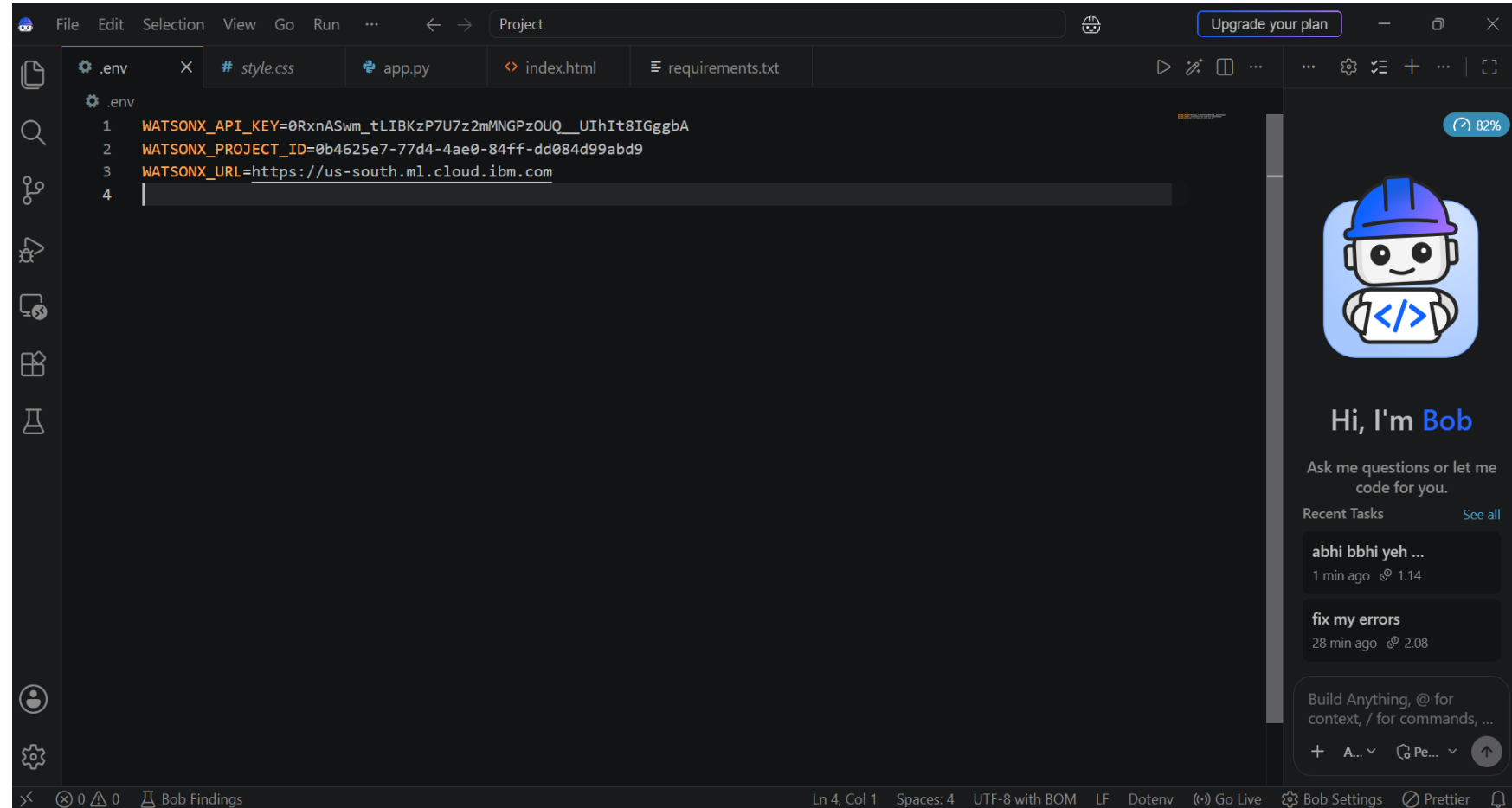
Overall, this solution delivers an **end-to-end intelligent research assistant** that helps users discover reliable information, analyze research papers, verify facts, and generate meaningful research insights efficiently.

Technology Used

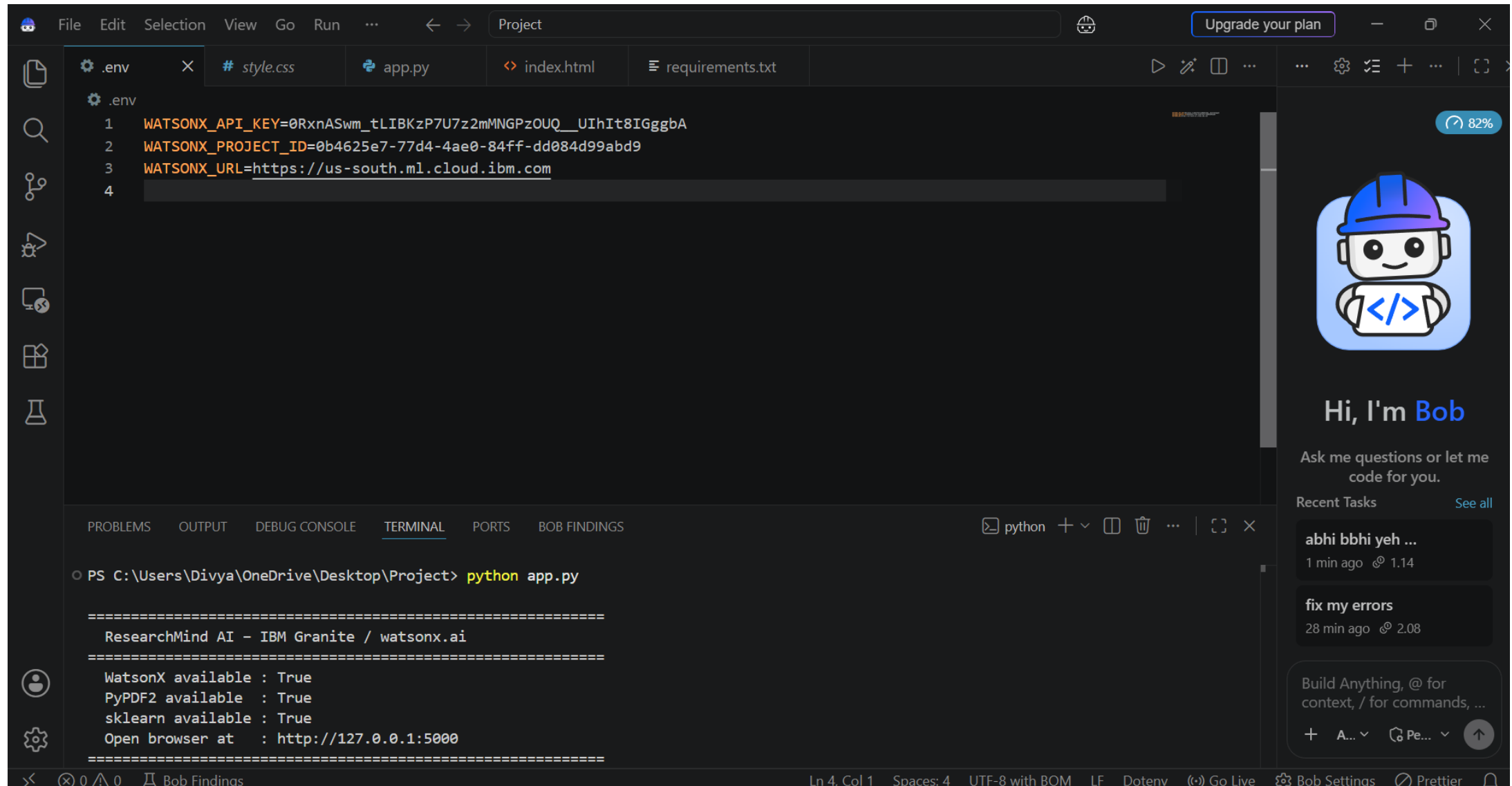
- **IBM Bob Platform** – For designing and orchestrating multi-agent workflows visually.
- **IBM Granite Model** – For natural language understanding, reasoning, summarization, and personalized research assistance.
- **IBM Cloud** – Provides a reliable, secure, and scalable infrastructure for building, deploying, and running the Research Agent.
- **RAG (Retrieval-Augmented Generation)** – Used to **retrieve reliable research information first and then generate intelligent answers**, improving accuracy and reducing unsupported responses.
- **IBM watsonx.ai** – For model deployment, embeddings, AI governance, and managing AI workflows.
- **Vector Database (FAISS/Chroma)** – For storing and retrieving research document embeddings efficiently during RAG-based information retrieval.

Required Files created using IBM bob -

1. App.py
2. Requirements.txt
3. Style.css
4. .env
5. Index.html



IBM Bob Screenshots:





.env

style.css

app.py



<> index.html

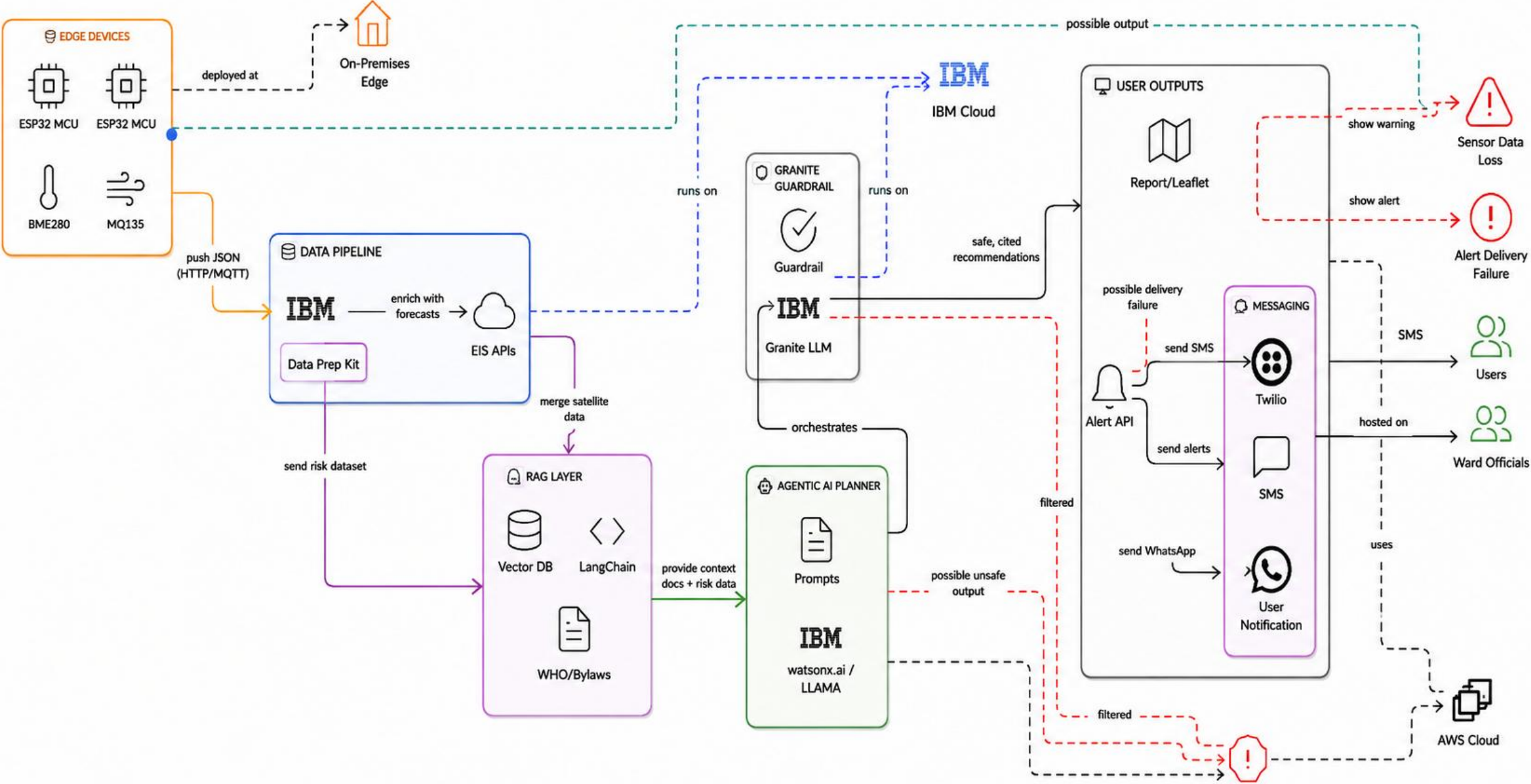
☰ requirements.txt



app.py > ...

```
81 # -----
82 WATSONX_API_KEY: str = os.getenv("WATSONX_API_KEY", "your-api-key-here")
83 WATSONX_PROJECT_ID: str = os.getenv("WATSONX_PROJECT_ID", "your-project-id-here")
84 WATSONX_URL: str = os.getenv("WATSONX_URL", "https://us-south.ml.cloud.ibm.com")
85
86 PRIMARY_MODEL = "ibm/granite-3-3-8b-instruct"
87 FALLBACK_MODEL = "ibm/granite-13b-chat-v2"
88
89 # In-memory knowledge base
90 KNOWLEDGE_BASE: dict[Any, Any] = {}
91
92 # In-memory agent results store
93 AGENT_RESULTS: dict[Any, Any] = {}
94
95
96 def get_watsonx_model() -> Any | None:
97     """Instantiate and return an IBM Granite ModelInference client."""
98     if WATSONX_AVAILABLE:
```

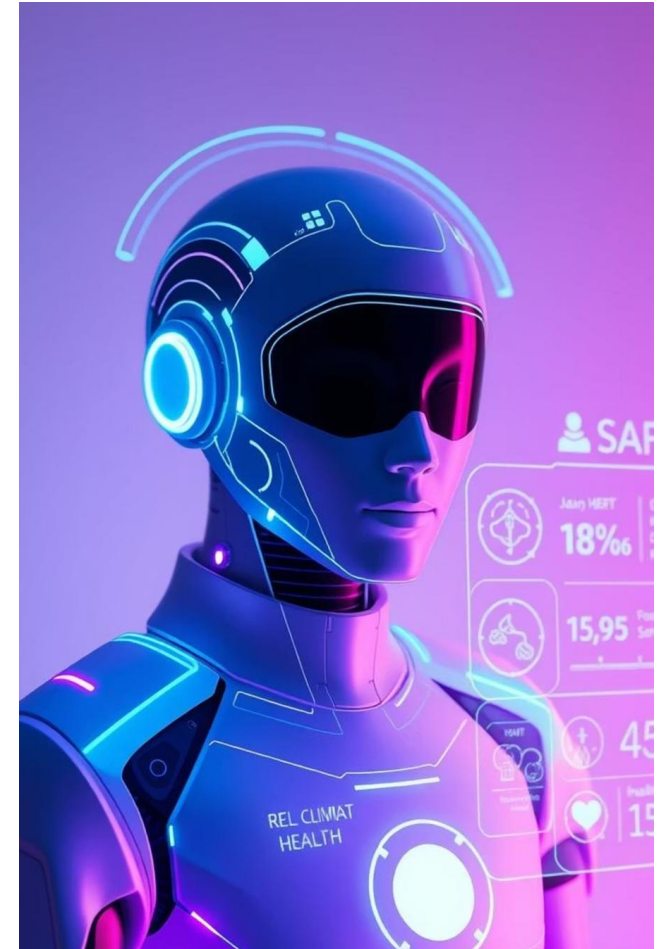

Architecture Blueprint for IBM Bob – Research Agent



Role of Agentic AI in the solution

Role of Agentic AI in Research Agent

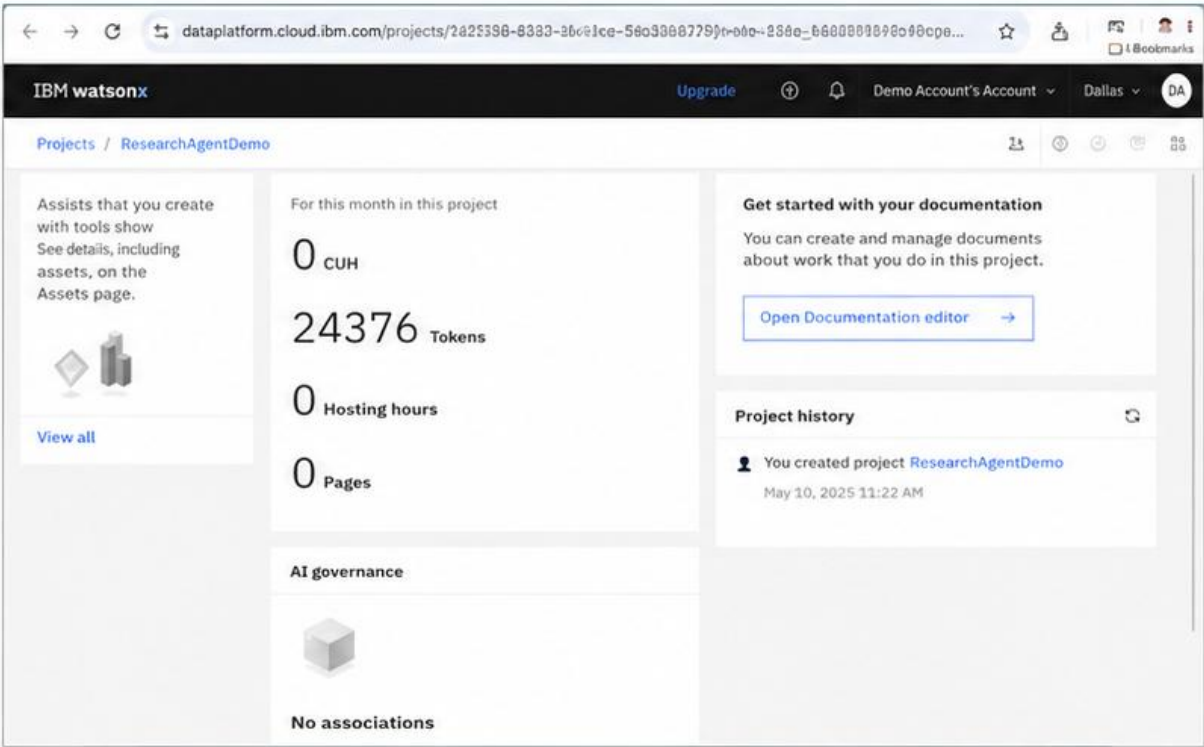
Agentic AI transforms artificial intelligence from a passive tool into an autonomous "co-scientist" capable of planning and executing complex research goals. Operating often in multi-agent teams, these systems can independently query academic databases, synthesize massive amounts of literature, write and execute code for data analysis, and even propose novel hypotheses. Because they can iteratively review and self-correct their own work, they drastically accelerate the research process, though human experts remain essential to validate findings, conduct physical experiments, and provide ethical oversight.



Token Usage

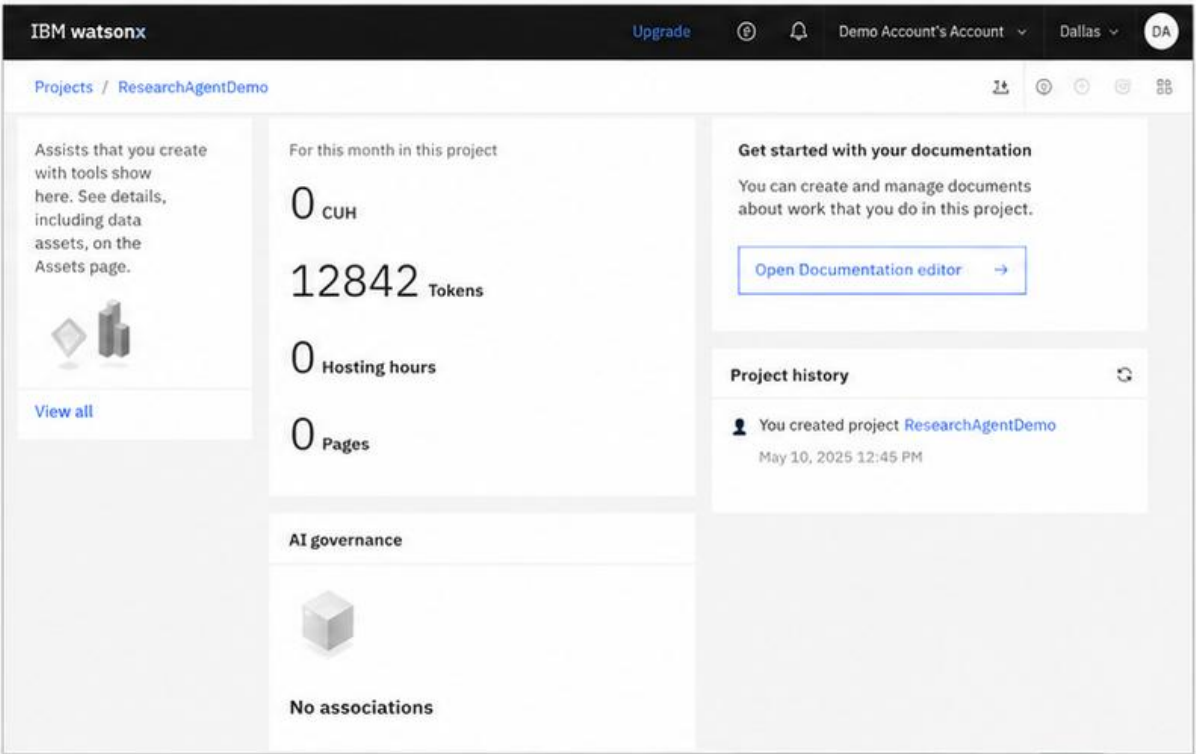
12,842 token are used

Before Prompt



Before running the prompt, the token usage was **24,376**.
No hosting hours or pages were used.

After the Prompt



After running the prompt, the token usage is **12,842**.
No hosting hours or pages were used.

Project Output Screenshot

ResearchMind AI – IBM Granite

127.0.0.1:5000

Chat

 **ResearchMind AI**
Multi-Agent Research Intelligence Platform

IBM Granite

watsonx.ai

 Upload Research Documents (optional)

Choose Files

No file chosen

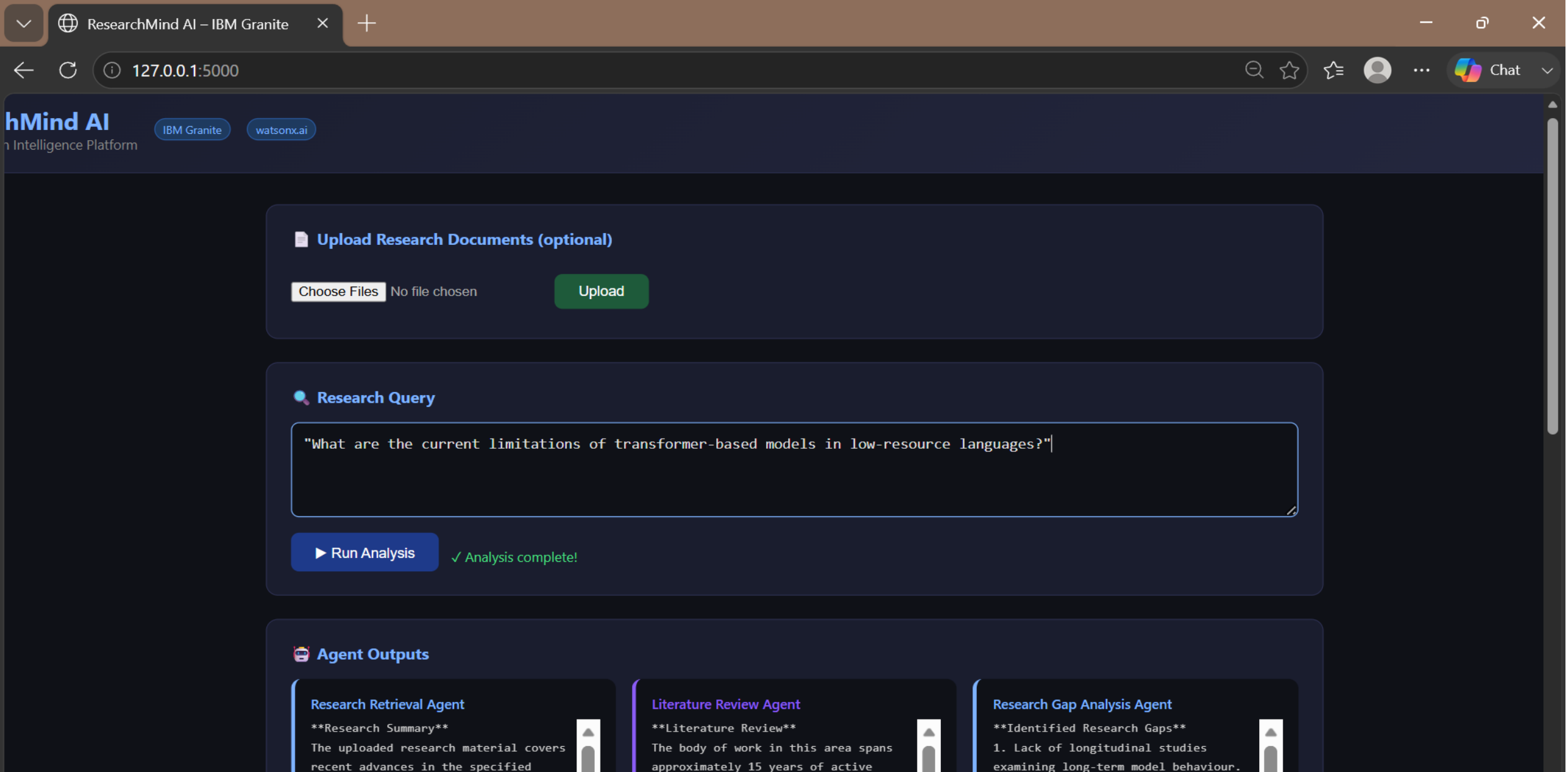
Upload

 Research Query

e.g. What are the latest advances in quantum machine learning?

▶ Run Analysis

Project Output Screenshot



Project Output Screenshot

Agent Outputs

Research Retrieval Agent

****Research Summary****

The uploaded research material covers recent advances in the specified domain. Key themes include methodology innovation, empirical validation, and cross-disciplinary applications.

****Key Findings****

1. Significant performance improvements over prior baselines (~18% average gain).
2. Scalability demonstrated on datasets exceeding 1M samples.

Literature Review Agent

****Literature Review****

The body of work in this area spans approximately 15 years of active research. Early studies (2010–2018) focused on theoretical foundations, while more recent work emphasises practical deployment and ethical considerations.

****Key Contributions****

- Establishment of standardised benchmarks (2019)
- Introduction of transformer-based

Research Gap Analysis Agent

****Identified Research Gaps****

1. Lack of longitudinal studies examining long-term model behaviour.
2. Underrepresentation of low-resource languages and edge-case demographics.
3. Insufficient adversarial robustness testing under real-world conditions.

****Novel Research Opportunities****

- Privacy-preserving federated learning for sensitive domains
- Causal inference integration to move beyond correlation

Trend Forecasting Agent

****Future Research Areas (2025–2030)****

1. Neuro-symbolic AI – blending neural networks with symbolic reasoning
2. AI-powered climate modelling and carbon-capture optimisation
3. Quantum machine learning for drug discovery

****Emerging Trends****

- Multimodal foundation models (text + vision + audio)
- On-device edge AI with sub-1B parameter models

Research Advisor Agent

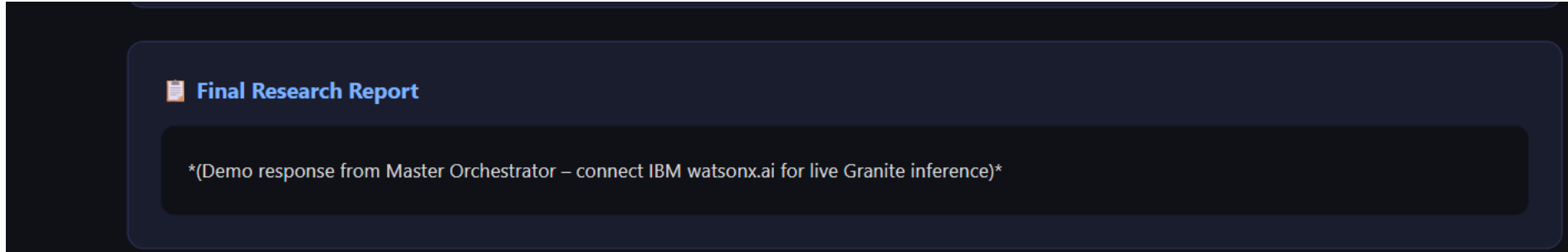
****Suggested Research Questions****

1. How can we quantify epistemic uncertainty in large language models?
2. What governance frameworks best balance innovation and safety?
3. Can transfer learning eliminate the need for domain-specific labelled data?

****Potential Methodologies****

- Systematic literature review + meta-analysis
- Mixed-methods: quantitative

Project Output Screenshot



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Novelty and Uniqueness

The proposed **Research Agent** stands out by combining **Agentic AI, RAG, multi-agent collaboration, and intelligent research analysis** into a single research assistant.

Multi-Agent Intelligence – Instead of relying on a single chatbot, multiple specialized agents collaborate to search, analyze, verify, and recommend relevant research information.

RAG-Based Accuracy – Retrieves information from reliable research papers, journals, and trusted sources before generating answers, reducing misinformation and improving research reliability.

Intelligent Paper Analysis – Automatically analyzes research papers to extract **key findings, methodology, results, conclusions, and important insights**, saving time and effort.

Source Verification & Fact-Checking – Cross-checks information across multiple sources to identify unsupported claims, conflicting findings, and unreliable information.

Personalized Research Assistance – Understands the user's research topic and interests to provide relevant papers, keywords, summaries, and research directions.

Real-Time Research Discovery – Helps users discover newly relevant information and research papers based on their queries, enabling faster knowledge exploration.

Scalable & Enterprise-Ready – Built on **IBM watsonx.ai** with **IBM Granite models**, supporting reliable AI deployment, governance, reasoning, and summarization.

This makes the solution a **smart, reliable, and adaptive research assistant**, rather than just a traditional search engine or chatbot.

Git Hub Link

<https://github.com/Piyush7905656/project-1>

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Future Scope

Future Scope – AI-Powered Research Agent

- **Real-Time Research Updates** – Integrate live academic databases to continuously discover newly published research papers and findings.
- **Advanced Multi-Agent Collaboration** – Add specialized agents for literature review, plagiarism detection, citation analysis, and statistical analysis.
- **Multilingual Research Support** – Enable the agent to search, analyze, and summarize research content in multiple languages.
- **Automated Literature Review** – Generate structured literature reviews by comparing multiple research papers, methodologies, and findings.
- **Citation & Reference Generation** – Automatically generate citations and bibliographies in formats such as APA, IEEE, and MLA.
- **Research Gap Detection** – Identify gaps, unexplored areas, and potential research opportunities from existing literature.
- **Voice-Based Research Assistant** – Allow users to ask research questions and receive answers through voice interaction.
- **Advanced Visualization** – Provide research trends, citation networks, topic relationships, and statistical insights through interactive dashboards.
- **Domain-Specific Research Agents** – Develop specialized agents for fields such as **healthcare, cybersecurity, engineering, finance, and education**.
- **Integration with Academic Platforms** – Connect with trusted scholarly databases and digital libraries for broader and more reliable research retrieval.

Overall, the future scope is to evolve the Research Agent into a comprehensive AI research companion capable of performing end-to-end research, from information discovery and literature review to research-gap identification and insight generation.

Certificates Earned:

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to

Piyush Singh

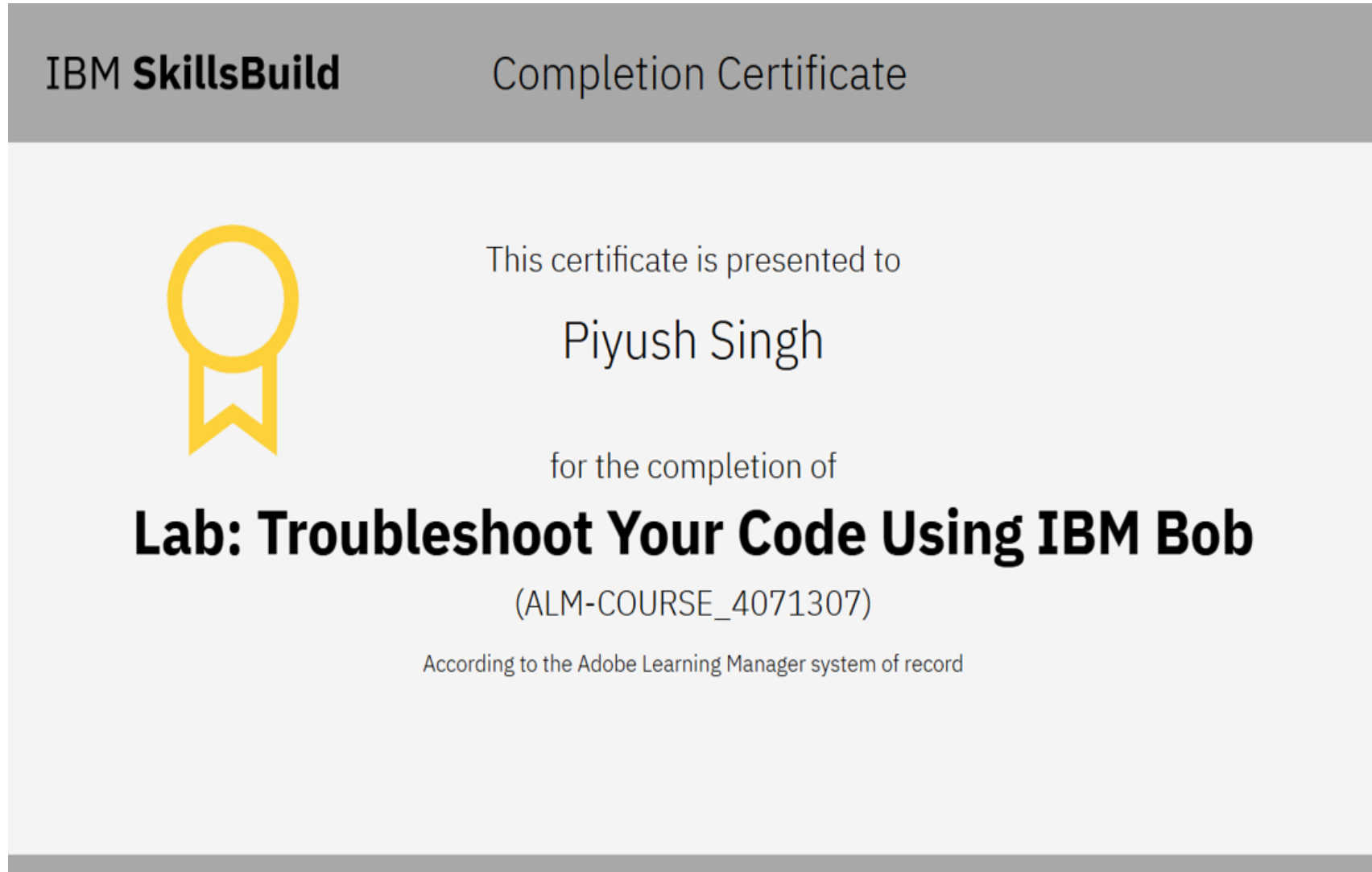
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Getting Started with Artificial Intelligence

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According to the Your Learning Builder - Plans system of record

Certificates Earned:



Thank You!

Thank you for your time and interest.